Upper Bound Energy Barriers for Simulated Annealing in Spherical Spin Glasses

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May 12, 2026

Abstract

We study upper bounds on the energy reached by finite-time Langevin dynamics and simulated annealing for spherical spin glass Hamiltonians. Motivated by Open Problem 1.11 of [HMRW25], which asks what energy simulated annealing obtains for mixed p-spin Hamiltonians, this report reviews the upper-bound mechanism in [Sel24b]'s proof for the pure spherical p-spin model and explains which part extends to mixed p-spin models. The proof proceeds by approximating hard spherical Langevin dynamics by a soft full-space dynamics, showing that the soft dynamics is a dimension-free Lipschitz function of the Gaussian disorder after conditioning on the Brownian path, and then invoking Lipschitz hardness. In the pure model, the relevant threshold is $E_\infty(p) = 2\sqrt{(p-1)/p}$. In the mixed model, the corresponding Lipschitz threshold is

$$\mathrm{ALG}(\xi) = \int_0^1 \sqrt{\xi''(q)}\,\mathrm{d}q.$$

We present the background, main results, and proof ideas needed to make the upper-bound argument accessible, and we explain why the matching lower-bound question for general mixed models remains open.

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1 Introduction

A central problem in high-dimensional probability is understanding the optimization of random objective functions: what values are attainable, and what algorithms achieve them? Unlike worst-case optimization, where convexity is often the main dividing line, random instances can exhibit sharp algorithmic thresholds and gaps between information-theoretic and computationally achievable optima. Spherical spin glass models [SK75, CS92, CK93, BADG06] provide a canonical setting for studying these phenomena. Their Hamiltonians are rugged random energy landscapes, making them useful models for high-dimensional nonconvex optimization, algorithmic barriers, and noisy dynamics. Simulated annealing is a natural algorithm in this setting: it starts at high temperature, where the dynamics explores broadly, and gradually increases the inverse temperature so that the chain favors higher-energy configurations. The central question is whether this procedure reaches near-optimal configurations or becomes trapped below the ground state at an algorithmic threshold.

We now defined the problem as follows: consider the probability space $(\Omega, \mathbb{P})$ where the state space is the high-dimensional sphere

$$\Omega = S_N = \sqrt{N} \cdot S^{N-1} := \{\sigma \in \mathbb{R}^N : \|\sigma\|_2 = \sqrt{N}\},$$

and we write $\mathbb{P} = \mathrm{Unif}(S_N)$ for the uniform probability measure on the sphere. The normalization $\|\sigma\|_2 = \sqrt{N}$ is standard in spherical spin glasses: it makes the Hamiltonian have fluctuations of order N, so that the natural quantity is the energy density $H_N(\sigma)/N$. Fix $p \geq 2$ and let $G_N^{(p)} \in (\mathbb{R}^N)^{\otimes p}$ be an order-p tensor with i.i.d. standard Gaussian entries $g_{i_1 \cdots i_p}^{(p)} \sim \mathcal{N}(0,1)$. The spherical *pure* p-spin Hamiltonian $H_{N,p} : S_N \to \mathbb{R}$ is the random homogeneous process

$$H_{N,p}(\sigma) := \frac{1}{N^{(p-1)/2}} \sum_{i_1,\ldots,i_p=1}^{N} g_{i_1 \cdots i_p}^{(p)} \sigma_{i_1} \cdots \sigma_{i_p} = N^{-(p-1)/2} \langle G_N^{(p)}, \sigma^{\otimes p} \rangle. \tag{1.1}$$

More generally, let $\{G_N^{(p)}\}_{p \geq 2}$ be independent Gaussian tensors, where each $G_N^{(p)} \in (\mathbb{R}^N)^{\otimes p}$ has i.i.d. standard Gaussian entries. Given coefficients $(\gamma_p)_{p \geq 2}$ with $\gamma_p \geq 0$, the spherical mixed p-spin Hamiltonian is

$$H_N(\sigma) := \sum_{p \geq 2} \gamma_p H_{N,p}(\sigma). \tag{1.2}$$

For simplicity, one may assume that the mixture is finite. More generally, one imposes standard summability assumptions on $(\gamma_p)_{p \geq 2}$ so that H_N and its first three derivatives are well-defined and uniformly controlled on $2\mathcal{B}_N$, where $\mathcal{B}_N := \{x \in \mathbb{R}^N : \|x\| \leq \sqrt{N}\}$. The mixture is encoded by $\xi(q) := \sum_{p \geq 2} \gamma_p^2 q^p$. For $\sigma_1, \sigma_2 \in S_N$, define the overlap function to be

$$R(\sigma_1, \sigma_2) := \frac{\langle \sigma_1, \sigma_2 \rangle}{N}.$$

Then H_N is a centered Gaussian process with covariance

$$\mathbb{E}\big[H_N(\sigma_1) H_N(\sigma_2)\big] = N \xi\big(R(\sigma_1, \sigma_2)\big).$$

The pure case is recovered by setting $\xi(x) = x^p$. This covariance formula is the main reason the model is analytically tractable. The random function H_N is not a collection of independent random values on the sphere; rather, two configurations have correlated energies according to their overlap. Configurations with $R(\sigma_1, \sigma_2) \approx 1$ are close on the sphere and have strongly correlated energies, while nearly orthogonal configurations have much weaker correlation. Thus the overlap is the basic geometric parameter controlling the landscape. For inverse temperature $\beta \geq 0$, the spherical Gibbs measure is

$$\mathrm{d}\mu_{\beta, H_N}(\sigma) = \frac{1}{Z_N(\beta)} \exp\big(\beta H_N(\sigma)\big)\, \mathrm{d}\mathbb{P}(\sigma),$$

where the partition function and free energy are

$$Z_N(\beta) := \int_{S_N} \exp\big(\beta H_N(\sigma)\big)\, \mathrm{d}\mathbb{P}(\sigma), \qquad F_N(\beta) := \frac{1}{N} \log Z_N(\beta). \tag{1.3}$$

At low temperature the Gibbs measure concentrates near high-energy configurations, so sampling from μ_{β,H_N} is closely related to optimizing H_N over the sphere. We use the Langevin normalization from [Sel24b]. In this convention, the parameter β in the SDE is the drift parameter; the corresponding Gibbs inverse temperature is 2β. Thus the spherical Langevin dynamics is

$$\mathrm{d}x_t = \left(\beta\,\nabla_{\mathrm{sp}} H_N(x_t) - \frac{N-1}{2N}\,x_t\right)\mathrm{d}t + P^{\perp}_{x_t}\,\mathrm{d}B_t. \tag{1.4}$$

Here $\nabla_{\mathrm{sp}} H_N(x) = P^{\perp}_x \nabla H_N(x)$ is the Riemannian gradient on S_N, $P^{\perp}_x$ is the orthogonal projection onto the tangent space $T_x S_N$, and B_t is standard Brownian motion in $\mathbb{R}^N$. The drift $\beta\nabla_{\mathrm{sp}} H_N$ pushes the process uphill in energy, while the projected noise keeps the dynamics on the sphere. With this normalization, the invariant measure is

$$\mathrm{d}\mu_{2\beta,H_N}(\sigma) \propto \exp(2\beta H_N(\sigma))\,\mathrm{d}\mathbb{P}(\sigma).$$

Therefore, when we say that the Langevin drift parameter is bounded by $\beta_{\max}$, the corresponding Gibbs inverse temperature is bounded by $2\beta_{\max}$.

Simulated annealing is the time-inhomogeneous version of this dynamics in which the Langevin drift parameter $\beta = \beta_t$ is slowly increased along a deterministic schedule. Informally, one starts from the uniform measure on S_N, runs the dynamics at a small inverse temperature, uses the endpoint as a warm start for a slightly larger inverse temperature, and repeats. The central question addressed in this work is what asymptotic energy this procedure reaches, and in particular whether it can exceed the Lipschitz algorithmic threshold $\mathrm{ALG}(\xi)$. The reason for studying simulated annealing is that it is one of the classical heuristics for nonconvex optimization. At high temperature, the noise dominates and the dynamics can move broadly through the energy landscape. At low temperature, the drift term biases the process toward higher-energy regions. The goal of simulted annealing and the annealing schedule is that by cooling slowly, the process avoids becoming trapped too early and eventually reaches near-optimal configurations. In spherical spin glasses, this heuristic can be tested against precise asymptotic thresholds such as the ground state energy and the Lipschitz algorithmic threshold.

There are two benchmark energy scales. The first is the information-theoretic optimum, or ground state energy,

$$\mathrm{OPT}^{\mathrm{Sp}}(\xi) := \operatorname*{p\text{-}lim}_{N\to\infty} \frac{1}{N} \max_{\sigma\in S_N} H_N(\sigma), \tag{1.5}$$

whenever the limit exists. This quantity is characterized by the zero-temperature limit of the Parisi formula. The second is the algorithmic threshold for dimension-free Lipschitz algorithms. For spherical mixed p-spin Hamiltonians, the works of [Sub21, EAMS21, Sel24a, HS25] identify this threshold as

$$\mathrm{ALG}(\xi) := \int_0^1 \sqrt{\xi''(q)}\,\mathrm{d}q. \tag{1.6}$$

Thus one generally expects

$$\mathrm{ALG}(\xi) \le \mathrm{OPT}^{\mathrm{Sp}}(\xi),$$

with equality only in special cases. This gap is one mathematical expression of the distinction between information-theoretic and algorithmic optimization. Motivated by Open Problem 1.11 of [HMRW25], this works asks where simulated annealing lies relative to these two thresholds.

Question. What energy does simulated annealing obtain when run on the mixed p-spin Hamiltonian (1.2)?

Equivalently, if σ_{SA} denotes the output of a specified annealing scheme, one would like to identify the limiting energy density

$$\mathrm{SA}(\xi) := \operatorname*{p\text{-}lim}_{N\to\infty} \frac{1}{N} H_N(\sigma_{\mathrm{SA}}), \tag{1.7}$$

assuming that this limit exists and is independent of the precise annealing schedule within a suitable class. For pure models, Sellke [Sel24b] resolves the corresponding energy question. The relevant threshold is

$$E_\infty(p) := 2\sqrt{\frac{p-1}{p}}. \tag{1.8}$$

Low-temperature Langevin dynamics reaches this value in the large-N limit, and this value is strictly below the Parisi ground state energy for $p \geq 3$. Moreover, $E_\infty(p)$ coincides with the Lipschitz algorithmic threshold in the pure case, since

$$\mathrm{ALG}(\xi)\big|_{\xi(q)=q^p} = \int_0^1 \sqrt{p(p-1)q^{p-2}}\,\mathrm{d}q = 2\sqrt{\frac{p-1}{p}} = E_\infty(p).$$

Thus, in the pure model, the dynamical threshold and the Lipschitz algorithmic threshold match.

1.1 Motivation and Results

This work focuses on the upper-bound mechanism in [Sel24b] proof and provides some original work for the mixed case. The generalized part of the upper bound argument has three steps: 1) approximate the hard spherical Langevin dynamics by a soft full-space dynamics, 2) show that the soft endpoint is a Lipschitz function of the disorder after conditioning on the Brownian path, and 3) invoke Lipschitz hardness. In the pure model this gives the threshold $E_\infty(p)$, while in the mixed model this same proof structure gives $\mathrm{ALG}(\xi)$.

Theorem 1.1 (Pure-model upper bound, [Sel24b])**.** *For any $p \geq 2$ and $\beta_0 \in [0, \infty)$, there exists $\delta = \delta(p, \beta_0) > 0$ such that the following holds for all $\beta \leq \beta_0$. Let $(x_t)_{t\in[0,T]}$ be spherical Langevin dynamics driven by the pure Hamiltonian $H_{N,p}$, initialized from $x_0 \in S_N$ independently of $H_{N,p}$. For any fixed $T \geq 0$, for N sufficiently large,*

$$\mathbb{P}\left[\sup_{t\in[0,T]} \frac{H_{N,p}(x_t)}{N} \leq E_\infty(p) - \delta\right] \geq 1 - e^{-cN}.$$

The proof has two layers. First, one approximates spherical Langevin dynamics by a dimension-free Lipschitz algorithm and obtains the weaker upper bound $E_\infty(p) + \eta$. Second, a pure-model gradient lower bound along the trajectory improves this to $E_\infty(p) - \delta$.

Theorem 1.2 (Mixed-model Lipschitz upper bound, informal)**.** *Let H_N be a mixed spherical p-spin Hamiltonian with covariance function*

$$\xi(q) = \sum_{p\geq 2} \gamma_p^2 q^p.$$

Fix $T < \infty$, $\beta_0 < \infty$, and $\eta > 0$. Let $(x_t)_{t\in[0,T]}$ be spherical Langevin dynamics with a deterministic inverse-temperature schedule $\beta_t \in [0, \beta_0]$, initialized from $x_0 \in S_N$ independently of H_N. Then, for N sufficiently large,

$$\mathbb{P}\left[\sup_{t\in[0,T]} \frac{H_N(x_t)}{N} \leq \mathrm{ALG}(\xi) + \eta\right] \geq 1 - e^{-cN},$$

where

$$\mathrm{ALG}(\xi) := \int_0^1 \sqrt{\xi''(q)}\,\mathrm{d}q.$$

This paper focuses only on the upper-bound side of the energy question. In the pure model, we review Sellke's proof of the bounded-temperature upper bound $E_\infty(p)-\delta$. For mixed models, we explain why the Lipschitz part of the argument remains valid after replacing $E_\infty(p)$ by $\mathrm{ALG}(\xi)$. We do not discuss the matching lower-bound argument, which in the pure model relies on approximate local maxima and saddle escape, and which remains open in this form for general mixed models.

2 Notation and Technical Preliminaries

We use the model notation from the Introduction throughout. Thus H_N denotes either a pure or mixed spherical spin glass Hamiltonian on S_N, with $H_{N,p}$ used when denoting a pure spherical spin glass Hamiltonian, and $\mathcal{B}_N = \{x \in \mathbb{R}^N : \|x\| \leq \sqrt{N}\}$ denotes the corresponding ball.

2.1 Geometry and Regularity

For a smooth function H_N defined near the sphere, ∇H_N and $\nabla^2 H_N$ denote the Euclidean gradient and Hessian. If $x \neq 0$, let $P_x^\perp$ be the orthogonal projection onto the orthogonal complement of x; for $x \in S_N$, which is the tangent space $T_x S_N$. The spherical gradient is

$$\nabla_{\mathrm{sp}} H_N(x) = P_x^\perp \nabla H_N(x).$$

We also use the radial derivative

$$\nabla_{\mathrm{rad}} H_N(x) := \frac{1}{N} \frac{\mathrm{d}}{\mathrm{d}t} H_N(tx)\Big|_{t=1},$$

and the corresponding Riemannian Hessian

$$\nabla^2_{\mathrm{sp}} H_N(x) := P_x^\perp \left(\nabla^2 H_N(x) - \nabla_{\mathrm{rad}} H_N(x) I_N \right) P_x^\perp.$$

These objects are especially important in comparing pure and mixed models: for a pure homogeneous Hamiltonian $\nabla_{\mathrm{rad}} H_N(x) = p H_N(x)/N$ on the sphere, while no such proportionality holds in a mixed model. For a multilinear tensor T of order k, we write

$$\|T\|_{\mathrm{op}} := \sup_{\|u_1\|=\cdots=\|u_k\|=1} \left| \langle T, u_1 \otimes \cdots \otimes u_k \rangle \right|.$$

For matrices this is the usual operator norm. Let

$$2\mathcal{B}_N := \{x \in \mathbb{R}^N : \|x\| \leq 2\sqrt{N}\}.$$

Following Sellke, we say that a three-times differentiable Hamiltonian $H_N : 2\mathcal{B}_N \to \mathbb{R}$ is C-bounded if

$$\|H_N\|_{W_N^{3,\infty}} := \sup_{x \in 2\mathcal{B}_N} \max\left(\frac{|H_N(x)|}{N}, \frac{\|\nabla H_N(x)\|}{\sqrt{N}}, \|\nabla^2 H_N(x)\|_{\mathrm{op}}, \sqrt{N}\, \|\nabla^3 H_N(x)\|_{\mathrm{op}} \right) \leq C.$$

This condition gives uniform control of the Hamiltonian and its first few derivatives on the region visited by the soft spherical dynamics.

2.2 Disorder Space and Lipschitz Algorithms

Let $\mathcal{H}_N$ denote the Hilbert space of disorder tensors defining H_N. In the pure p-spin case this is $(\mathbb{R}^N)^{\otimes p}$ with coefficient-wise L^2 norm: if H_N and H'_N have tensor entries $(g_{i_1,\ldots,i_p})$ and $(g'_{i_1,\ldots,i_p})$, then

$$\|H_N - H'_N\|^2_{\mathcal{H}_N} = \sum_{i_1,\ldots,i_p=1}^{N} \left| g_{i_1,\ldots,i_p} - g'_{i_1,\ldots,i_p} \right|^2.$$

For a finite mixed model, $\mathcal{H}_N$ is the direct sum over the degrees appearing in the mixture, again with coefficient-wise L^2 norm. Thus distances in $\mathcal{H}_N$ measure perturbations of the disorder. An L-Lipschitz optimization algorithm is a map $\mathcal{A} : \mathcal{H}_N \to \mathcal{B}_N$ satisfying

$$\|\mathcal{A}(H_N) - \mathcal{A}(H'_N)\| \leq L \|H_N - H'_N\|_{\mathcal{H}_N}.$$

This captures stable algorithms: small perturbations of the random tensor cannot produce large jumps in the output. We use the same notion for randomized algorithms. If $\mathcal{A} : \mathcal{H}_N \times \mathcal{U} \to \mathcal{B}_N$ has external randomness U independent of H_N, then $\mathcal{A}$ is L-Lipschitz almost surely if, for almost every fixed u, the map $H_N \mapsto \mathcal{A}(H_N; u)$ is L-Lipschitz. The relevant hardness input is that dimension-free Lipschitz algorithms cannot exceed the algorithmic threshold: in the pure model this threshold is $E_\infty(p)$, and in the mixed model it is $\mathrm{ALG}(\xi)$.

The phrase *dimension-free* means that the Lipschitz constant may depend on fixed parameters of the model, such as p, ξ, T, or $\beta_{\max}$, but not on the dimension N. This distinction is important: a bound with Lipschitz constant growing with N would not be strong enough to invoke the Lipschitz-hardness theorem in the large-N limit. In the Langevin application, the algorithm is randomized because it depends on the Brownian path. The key reduction is to condition on the Brownian path $B_{[0,T]}$. Once this path is fixed, the soft Langevin endpoint becomes a deterministic map of the disorder. If this deterministic map is dimension-free Lipschitz for almost every Brownian path, then the randomized algorithm is also covered by the Lipschitz-hardness result after averaging over the Brownian randomness.

2.3 Annealed Markov Chains and Simulated Annealing

We now recall the general annealing framework. Let H be any Hamiltonian on a state space Ω, and let $(\mu_{\beta,H})_{\beta\geq 0}$ be Gibbs measures

$$\mathrm{d}\mu_{\beta,H}(\sigma) = \frac{1}{Z(\beta)} e^{\beta H(\sigma)}\, \mathrm{d}\rho(\sigma),$$

where ρ is a reference measure. For each inverse temperature β, let $P_{\beta H}$ be a reversible and ergodic Markov chain with stationary distribution $\mu_{\beta,H}$. An annealing scheme is a time-inhomogeneous chain obtained by running $P_{\beta H}$ while slowly increasing β.

2.4 Two Stability Tools

We will use two standard stability tools in the upper-bound argument. The first is Gronwall's inequality. In the form needed here, if a nonnegative function W_t satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} W_t \leq a W_t + b$$

on $[0, T]$, then

$$W_T \leq e^{aT} W_0 + \frac{b}{a}(e^{aT} - 1).$$

In particular, when $W_0 = 0$ and b is proportional to the squared distance between two disorder tensors, Gronwall's inequality converts a differential stability estimate into a Lipschitz bound for the endpoint of the dynamics.

The second is Kirszbraun's extension theorem. Since the disorder space $\mathcal{H}_N$ and the ambient space $\mathbb{R}^N$ are Hilbert spaces, any L-Lipschitz map from a subset $K \subseteq \mathcal{H}_N$ into $\mathbb{R}^N$ extends to an L-Lipschitz map on all of $\mathcal{H}_N$. We use this because the Langevin endpoint is first shown to be Lipschitz only on the high-probability set K_N of C-bounded Hamiltonians. Kirszbraun's theorem lets us extend this map to the full disorder space, so that the Lipschitz-hardness theorem applies.

Definition 1 (Annealing Scheme)**.** Let H be a Hamiltonian on a state space Ω, let ρ be a reference measure on Ω, and let $(\mu_\beta)_{\beta\geq 0}$ be the associated Gibbs measures

$$\mathrm{d}\mu_\beta(\sigma) = \frac{1}{Z(\beta)} e^{\beta H(\sigma)}\, \mathrm{d}\rho(\sigma).$$

For each $\beta \leq \beta_0$, let P_β be a reversible and ergodic Markov chain with stationary distribution μ_β. An *inverse-temperature schedule* is a function

$$\beta : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}.$$

An *annealing scheme* $\mathcal{A}$ is the time-inhomogeneous Markov chain which, at time t, applies the transition kernel $P_{\beta(t)}$.

Thus high temperature corresponds to small β, where the chain explores the state space broadly, while low temperature corresponds to large β, where the Gibbs measure increasingly favors high-energy configurations. The purpose of annealing is to increase β slowly enough that the chain approximately tracks the family of Gibbs measures $(\mu_\beta)_{\beta\geq 0}$.

Definition 2 (Simulated Annealing)**.** In the normalized discrete-time schedule, set

$$\beta_i = \frac{i}{T}, \qquad i = 0, \ldots, T.$$

The algorithm initializes from the high-temperature distribution $\mu_0 = \rho$. Then, for $i = 1, \ldots, T$, the i-th stage runs the Markov chain P_{β_i} for $\mathrm{poly}(N)$ steps, initialized at the output of the previous stage.

More generally, one may replace the linear schedule $\beta_i = i/T$ by any monotone schedule

$$0 = \beta_0 < \beta_1 < \cdots < \beta_T \leq \beta_{\max}.$$

The essential feature is the warm-start structure: the endpoint of the chain at inverse temperature β_i is used as the initialization for the next stage at inverse temperature β_{i+1}. For spherical spin glasses, the natural continuous-time analogue of P_β is Langevin diffusion on S_N. The annealed Langevin dynamics is obtained by replacing the constant inverse temperature parameter by a deterministic schedule. Thus, for a piecewise Lipschitz schedule

$$\beta_t : [0, T] \to [0, \beta_{\max}],$$

we consider

$$\mathrm{d}Z_t = \left(\beta_t \nabla_{\mathrm{sp}} H_N(Z_t) - \frac{N-1}{2N} Z_t\right) \mathrm{d}t + P^{\perp}_{Z_t} \, \mathrm{d}B_t. \tag{2.1}$$

This is the continuous-time version of simulated annealing. In this paper the time horizon $T < \infty$ is fixed independently of N. This restriction is important: the Lipschitz constant obtained from the Gronwall stability argument depends on T, so the finite-time reduction does not immediately apply to polynomially long annealing schedules.

2.5 Why Annealed Langevin Can Be Compared to Lipschitz Algorithms

The upper-bound argument uses a stability principle: over a fixed finite time horizon, annealed Langevin dynamics can be approximated by an algorithm whose output depends Lipschitz-continuously on the disorder. The hard spherical process

$$\mathrm{d}Z_t = \left(\beta_t \nabla_{\mathrm{sp}} H_N(Z_t) - \frac{N-1}{2N} Z_t\right) \mathrm{d}t + P^{\perp}_{Z_t} \, \mathrm{d}B_t$$

is not immediately Lipschitz in H_N, because the Brownian noise is projected onto the tangent space at the current location. Thus, if two Hamiltonians H_N and $\widetilde{H}_N$ are driven by the same Brownian motion, the noise terms are still different:

$$P^{\perp}_{Z_t} \, \mathrm{d}B_t \qquad \text{and} \qquad P^{\perp}_{\widetilde{Z}_t} \, \mathrm{d}B_t.$$

Sellke's proof resolves this by replacing the hard spherical dynamics with a soft full-space dynamics in $\mathbb{R}^N$. The soft process uses a radial penalty to keep the trajectory close to S_N, but its Brownian noise is unprojected. Hence, after conditioning on the Brownian path and initialization, the endpoint becomes a deterministic map of the disorder,

$$H_N \mapsto Y_T(H_N).$$

On the high-probability event that H_N is C-bounded, the derivative bounds in the $W_N^{3,\infty}$ norm imply, via Gronwall's inequality, that this map is L_T-Lipschitz, where L_T depends on T, the temperature schedule, and the regularity constant, but not on N. Thus finite-horizon annealed Langevin dynamics can be compared to a randomized Lipschitz algorithm, up to the error from the soft approximation. Section 3 carries out this reduction and applies the corresponding Lipschitz hardness bound.

3 Upper Bound: Proof Sketch for Theorem 1.1

The argument below sketches the Lipschitz part of the pure-model upper bound. This part gives the weaker estimate

$$\sup_{t \in [0,T]} \frac{H_{N,p}(x_t)}{N} \le E_\infty(p) + \eta$$

with high probability. Sellke then upgrades this to the stated $E_\infty(p) - \delta$ bound using an additional gradient lower-bound refinement. We summarize this final step only briefly, since the Lipschitz mechanism is the part that extends directly to mixed models. Recall that for the pure spherical p-spin model, the Lipschitz algorithmic threshold is $E_\infty(p) = 2\sqrt{\frac{p-1}{p}}$. The pure upper-bound proof has two layers. First, the soft spherical approximation and Lipschitz reduction show that finite-horizon Langevin dynamics cannot exceed $E_\infty(p) + \eta$ with high probability. Second, a pure-model gradient lower-bound argument improves this to $E_\infty(p) - \delta$, giving Theorem 1.1.

3.1 Soft Spherical Approximation

The first obstacle in applying a Lipschitz-hardness result to the Langevin trajectory x_t is that x_t lives on the sphere S_N, where the Brownian increment is projected onto the tangent space:

$$P_{x_t}^{\perp}\,\mathrm{d}B_t.$$

Thus, two trajectories driven by the same ambient Brownian motion but in two different Hamiltonians receive different projected noise inputs. To obtain a stable function of the disorder, the first step is to replace the hard spherical constraint by a soft radial constraint. Let

$$q(y) := \frac{\|y\|^2}{N}.$$

Following Sellke, define the soft radial potential as a function of the squared radius ratio q:

$$f_L(q) := L(q-1)^2 + (q^2-1)^p.$$

For L sufficiently large, consider the full-space soft dynamics $y_t^{(L)}$ on $\mathbb{R}^N$, initialized from $y_0^{(L)} = x_0 \in S_N$, and solving

$$\mathrm{d}y_t^{(L)} = \Big(\beta_t \nabla H_{N,p}(y_t^{(L)}) - f_L'\Big(q(y_t^{(L)})\Big)\, y_t^{(L)}\Big)\,\mathrm{d}t + \mathrm{d}B_t.$$

Here $\beta_t \in [0, \beta_{\max}]$ is a deterministic piecewise Lipschitz schedule. The diffusion term is now the unprojected Brownian motion $\mathrm{d}B_t$, so two soft trajectories driven by the same B_t but different Hamiltonians receive identical noise inputs. The price is that $y_t^{(L)}$ no longer lies exactly on S_N. The radial term $-f_L'(q(y_t^{(L)}))y_t^{(L)}$ is a confining force which dominates near $q = 1$ by the $L(q-1)^2$ part of the potential, which pulls the process back toward the sphere.

Lemma 3.1 (Soft spherical approximation)**.** *Fix $x_0 \in S_N$ and let $H_{N,p} : \mathbb{R}^N \to \mathbb{R}$ be C-bounded. For every $\varepsilon > 0$, $\beta_{\max} < \infty$, and $T < \infty$, there exist $L, c > 0$ such that the following holds for all sufficiently large N. Let x_t be the hard spherical Langevin dynamics and $y_t^{(L)}$ the soft dynamics above, driven by the same Brownian motion and initialized from $x_0 = y_0^{(L)} \in S_N$. Then*

$$\mathbb{P}\left[\sup_{t\in[0,T]} \|x_t - y_t^{(L)}\| \le \varepsilon\sqrt{N}\right] \ge 1 - e^{-cN}.$$

The same conclusion holds uniformly over deterministic piecewise Lipschitz schedules $\beta_t : [0,T] \to [0, \beta_{\max}]$.

Proof Sketch. The proof has two parts. First, one controls the radial deviation of the soft process. Define$z_t := \Big(N^{-1/2}\|y_t^{(L)}\| - 1\Big)^2$. The confining potential forces z_t to remain small. Up to the first time at which z_t exceeds a fixed threshold, Itô's formula gives a drift bound of the form$b_t \le -2Lz_t + O_{\beta_{\max}}(1)\sqrt{z_t}$. For L large, the negative confining term dominates. Applying an exponential supermartingale argument to $M_t = \exp(\eta L z_t N)$, for a small constant $\eta > 0$, andusing optional stopping gives $\sup_{t\in[0,T]} z_t \le \delta$ with probability at least $1 - e^{-cN}$. Equivalently, the soft trajectory remains within distance $O(\sqrt{\delta N})$ of the sphere.Second, one compares the hard spherical trajectory x_t with the normalized soft trajectory

$$\widetilde{y}_t^{(L)} := \frac{\sqrt{N}\, y_t^{(L)}}{\|y_t^{(L)}\|}.$$

Since z_t is small, the normalization changes the coefficients of the soft dynamics only by a small amount, so $\widetilde{y}_t^{(L)}$ satisfies almost the same SDE as x_t on the sphere. To quantify this comparison, set$w_t := N^{-1}\|x_t - \widetilde{y}_t^{(L)}\|^2$. By Itô's formula and the$C$-boundedness of $H_{N,p}$, the drift of w_t satisfies the self-bounding estimate $b_t \le C(\beta_{\max}+1)(w_t+\delta)$. A Gronwall-type argument then gives$\sup_{t\in[0,T]} w_t \le e^{C\beta_{\max}T}O(\delta)$ with probability atleast $1 - e^{-cN}$. Choosing$\delta = \delta(\varepsilon, \beta_{\max}, T)$ sufficiently small yields

$$\sup_{t\in[0,T]} \|x_t - \widetilde{y}_t^{(L)}\| \le \varepsilon\sqrt{N}/2.$$

Combining this estimate with the radial estimate for $y_t^{(L)}$, we obtain

$$\sup_{t\in[0,T]} \|x_t - y_t^{(L)}\| \le \varepsilon\sqrt{N},$$

which proves the desired soft spherical approximation. □

3.2 Lipschitz Approximation

Once the soft dynamics $y_t^{(L)}$ has been constructed, the next step is to view the endpoint $y_T^{(L)}$ as a function of the disorder $H_{N,p}$, with the Brownian path $B_{[0,T]}$ fixed as a frozen source of randomness. After conditioning on $B_{[0,T]}$, the soft SDE is deterministic in $H_{N,p}$. The key point is that this deterministic map is Lipschitz.

Lemma 3.2 (Lipschitz approximation)**.** *For L sufficiently large and fixed x_0, there exists, for almost every Brownian path $B_{[0,T]}$, an $O_{T,\beta_{\max},L}(1)$-Lipschitz map*

$$\ell^{(L)}_{B_{[0,T]}} : \mathcal{H}_N \to \mathcal{B}_N$$

such that

$$\mathbb{P}\left[(1+\varepsilon)\ell^{(L)}_{B_{[0,T]}}(H_{N,p}) = y_T^{(L)}\right] \ge 1 - e^{-cN}.$$

The same conclusion holds uniformly over deterministic piecewise Lipschitz schedules $\beta_t : [0,T] \to [0,\beta_{\max}]$.

Proof Sketch. Fix the Brownian path $B_{[0,T]}$. We show that the map $H_{N,p} \mapsto y_T^{(L)}$ is Lipschitz on the set K_N of C-bounded Hamiltonians, up to the high-probability event on which the soft trajectory stays close to the sphere. Let $H^{(1)}_{N,p}, H^{(2)}_{N,p} \in K_N$, and let $y^{(L)}_{t,1}$ and $y^{(L)}_{t,2}$ be the corresponding soft trajectories driven by the same Brownian motion and initialized from the same point x_0. Set

$$W_t := \|y^{(L)}_{t,1} - y^{(L)}_{t,2}\|^2.$$

Since the diffusion coefficient in the soft dynamics is the identity, the Brownian increments cancel when we subtract the two SDEs. Applying Itô's formula to

$$W_t := \|y^{(L)}_{t,1} - y^{(L)}_{t,2}\|^2,$$

the only remaining terms come from the difference of the drifts. The C-boundedness of the Hamiltonians and the smoothness of the confining potential give

$$\frac{1}{2}\,\mathrm{d}W_t \le O_{\beta_{\max},L}\left(W_t + \|H^{(1)}_{N,p} - H^{(2)}_{N,p}\|^2_{\mathcal{H}_N}\right)\mathrm{d}t.$$

Since $W_0 = 0$, Gronwall's inequality yields

$$W_T \le O_{T,\beta_{\max},L}\|H^{(1)}_{N,p} - H^{(2)}_{N,p}\|^2_{\mathcal{H}_N}.$$

Thus, for almost every fixed Brownian path, the map $H_{N,p} \mapsto y_T^{(L)}$ is $O_{T,\beta_{\max},L}(1)$-Lipschitz on the C-bounded set K_N.

To apply Lipschitz hardness, the algorithm must be defined on all of $\mathcal{H}_N$, not only on K_N. By Kirszbraun's extension theorem, the map defined on K_N extends to an $O_{T,\beta_{\max},L}(1)$-Lipschitz map on the full disorder space without increasing its Lipschitz constant. After the harmless rescaling by $(1+\varepsilon)^{-1}$, this gives a map

$$\ell^{(L)}_{B_{[0,T]}} : \mathcal{H}_N \to \mathcal{B}_N$$

such that, on the high-probability event where $H_{N,p} \in K_N$ and the soft approximation holds,

$$(1+\varepsilon)\ell^{(L)}_{B_{[0,T]}}(H_{N,p}) = y_T^{(L)}.$$

Therefore the soft Langevin endpoint can be treated as the output of a randomized dimension-free Lipschitz algorithm applied to $H_{N,p}$. □

3.3 Application of Lipschitz Hardness

We can now invoke the Lipschitz-hardness theorem. For the pure p-spin model, any dimension-free Lipschitz algorithm $\mathcal{A} : \mathcal{H}_N \to \mathcal{B}_N$ satisfies, for every fixed $\eta > 0$,

$$\mathbb{P}\left[\frac{H_{N,p}(\mathcal{A}(H_{N,p}))}{N} \geq E_\infty(p) + \eta\right] \leq e^{-cN}.$$

Conditioning on the Brownian path, the map $\ell^{(L)}_{B_{[0,T]}}$ is such a Lipschitz algorithm, and averaging over the Brownian path gives the same high-probability bound for $\ell^{(L)}_{B_{[0,T]}}(H_{N,p})$.

It remains to transfer this estimate from the rescaled ball-valued output $\ell^{(L)}_{B_{[0,T]}}(H_{N,p})$ to the soft endpoint $y^{(L)}_T$. On the high-probability event from the Lipschitz approximation lemma,

$$y^{(L)}_T = (1+\varepsilon)\ell^{(L)}_{B_{[0,T]}}(H_{N,p}).$$

In the pure case, homogeneity gives

$$H_{N,p}\big((1+\varepsilon)u\big) = (1+\varepsilon)^p H_{N,p}(u).$$

Choosing $\varepsilon > 0$ sufficiently small compared with η, the energy change caused by this rescaling is absorbed into the ηN error. Combining this with the soft approximation yields

$$\frac{H_{N,p}(x_T)}{N} \leq E_\infty(p) + \eta$$

with high probability. To make the bound simultaneous over $t \in [0,T]$, one applies the same argument on a fine deterministic time net and then uses a short-time movement estimate for Langevin dynamics to pass from the net to the whole interval.

3.4 The Final Pure-Model Refinement

The Lipschitz argument gives the upper bound $E_\infty(p)+\eta$. Sellke's pure-model theorem strengthens this to $E_\infty(p)-\delta$ using an additional input specific to pure p-spin Hamiltonians: along any bounded temperature Langevin trajectory, the spherical gradient remains bounded below with high probability. Informally, for some $\eta_0 = \eta_0(p, \beta_{\max}) > 0$,

$$N^{-1/2}\|\nabla_{\mathrm{sp}} H_{N,p}(x_t)\| \geq \eta_0$$

throughout any fixed time interval. The refinement is a contradiction argument. Suppose that at some time t, the trajectory reaches energy very close to $E_\infty(p)$. Restart the dynamics from x_t using a suitably chosen bounded inverse temperature. The gradient lower bound then forces a definite increase in energy over a short time interval, say

$$H_{N,p}(\widetilde{x}_{t+s}) - H_{N,p}(x_t) \geq 2\delta N$$

with high probability. But the restarted dynamics is still a bounded temperature finite-time Langevin procedure, so the Lipschitz upper bound from the previous subsection applies to it as well. Therefore the restarted trajectory cannot exceed $E_\infty(p) + \delta$. Combining the two inequalities forces

$$\frac{H_{N,p}(x_t)}{N} \leq E_\infty(p) - \delta.$$

A time-net argument then makes this bound uniform over $t \in [0,T]$, proving Theorem 1.1.

4 Upper Bound: The Mixed Model and the Threshold $\mathrm{ALG}(\xi)$

We now explain how the Lipschitz part of the pure upper-bound argument extends to mixed spherical p-spin Hamiltonians. Throughout this section, H_N denotes the mixed Hamiltonian with covariance function $\xi(q) = \sum_{p\geq 2} \gamma_p^2 q^p$. The purpose of this section is not to prove a complete theorem for simulated annealing in mixed models. Rather, we explain the part of the pure proof that is robust: for fixed time horizon $T < \infty$, bounded deterministic drift schedule $\beta_t \in [0, \beta_{\max}]$, and fixed error $\eta > 0$, the soft-approximation and Lipschitz-hardness mechanism gives the finite-horizon upper bound

$$\mathbb{P}\left[\sup_{t\in[0,T]} \frac{H_N(x_t)}{N} \leq \mathrm{ALG}(\xi) + \eta\right] \geq 1 - e^{-cN},$$

for all sufficiently large N. Here x_t denotes the annealed spherical Langevin dynamics driven by H_N, and

$$\mathrm{ALG}(\xi) = \int_0^1 \sqrt{\xi''(q)}\,\mathrm{d}q.$$

This is the mixed analogue of the $E_\infty(p) + \eta$ Lipschitz layer in the pure proof.

4.1 The Mixed Lipschitz-Hardness Input

The key external input is the mixed-model Lipschitz-hardness theorem, which says that dimension-free Lipschitz algorithms cannot exceed the algorithmic threshold $\mathrm{ALG}(\xi)$. More precisely, for a mixed Hamiltonian satisfying the usual assumptions on ξ, and for any dimension-free Lipschitz map $\mathcal{A}_N : \mathcal{H}_N \to \mathcal{B}_N$, one has, for every fixed $\eta > 0$,

$$\mathbb{P}\left[\frac{H_N(\mathcal{A}_N(H_N))}{N} \geq \mathrm{ALG}(\xi) + \eta\right] \leq e^{-cN}.$$

This is the mixed analogue of the pure-model hardness statement with $E_\infty(p)$ replaced by $\mathrm{ALG}(\xi)$; see [Sub21, EAMS21, Sel24a, HS25] for the Lipschitz-algorithmic threshold picture. The important point for us is that the theorem applies to algorithms whose Lipschitz constants are bounded independently of the dimension N (stable algorithms).

4.2 Extension of the Soft Approximation

The soft-approximation step can also be used for the mixed model. In the pure proof, the hard spherical dynamics is compared to a full-space soft dynamics with a coercive radial penalty. For mixed Hamiltonians, the same framework applies provided the Hamiltonian is C-bounded on the relevant ball. For finite mixtures, this follows by applying the pure derivative estimates degree by degree and taking a union bound over the finitely many degrees appearing in the mixture. For infinite mixtures, one imposes the standard summability assumptions on $(\gamma_p)_{p\geq 2}$, ensuring that H_N and its first few derivatives are uniformly controlled on $2\mathcal{B}_N$. Under these assumptions, one may use the same kind of coercive radial potential as in the pure proof. The exact formula of the radial potential is not the main issue; what matters is that its radial drift keeps the soft trajectory within $o(\sqrt{N})$ of the sphere over every fixed time interval. Thus, if $y_t^{(L)}$ denotes the soft full-space dynamics driven by the same Brownian motion as x_t, then for L sufficiently large,

$$\sup_{t\in[0,T]} \|x_t - y_t^{(L)}\| \leq \varepsilon\sqrt{N}$$

with high probability. This is the mixed-model version of the soft spherical approximation.

4.3 The Lipschitz Reduction

After conditioning on the Brownian path $B_{[0,T]}$, the soft dynamics is a deterministic function of the disorder. If $H_N^{(1)}$ and $H_N^{(2)}$ are two C-bounded mixed Hamiltonians and $y_{t,1}^{(L)}, y_{t,2}^{(L)}$ are the corresponding soft trajectories driven by the same Brownian path, then subtracting the two soft SDEs cancels the Brownian noise. The remaining difference is

controlled by the difference between the two Hamiltonians and by the Lipschitz dependence of the drift. A Gronwall estimate gives

$$\|y_{T,1}^{(L)} - y_{T,2}^{(L)}\| \le O_{T,\beta_{\max},L,\xi}(1)\|H_N^{(1)} - H_N^{(2)}\|_{\mathcal{H}_N}.$$

Thus, for fixed T, the soft endpoint is a dimension-free Lipschitz function of the mixed disorder on the C-bounded set. As in the pure proof, Kirszbraun's theorem extends this Lipschitz map to all of $\mathcal{H}_N$ without increasing its Lipschitz constant. Therefore, after conditioning on the Brownian path, we obtain a dimension-free Lipschitz algorithm

$$\ell_{B_{[0,T]}}^{(L)} : \mathcal{H}_N \to \mathcal{B}_N.$$

Applying the mixed Lipschitz-hardness theorem gives

$$\frac{H_N(\ell_{B_{[0,T]}}^{(L)}(H_N))}{N} \le \mathrm{ALG}(\xi) + \eta$$

with high probability, after averaging over the Brownian path. It remains to transfer this estimate from the rescaled ball-valued Lipschitz output back to the soft endpoint. On the high-probability event from the Lipschitz approximation step, the soft endpoint can be written as

$$y_T^{(L)} = (1+\varepsilon)\ell_{B_{[0,T]}}^{(L)}(H_N).$$

Unlike the pure model, the mixed model has no single homogeneity degree. Instead, we use the derivative bounds on H_N over $2\mathcal{B}_N$. For $u \in \mathcal{B}_N$ and $\varepsilon > 0$ small,

$$\frac{|H_N((1+\varepsilon)u) - H_N(u)|}{N} \le C_\xi \varepsilon.$$

Choosing ε sufficiently small relative to η, this rescaling error is absorbed into the final ηN tolerance. Hence

$$\frac{H_N(y_T^{(L)})}{N} \le \mathrm{ALG}(\xi) + \eta$$

with high probability. The soft approximation then transfers the estimate from $y_T^{(L)}$ to the hard spherical trajectory x_T. Finally, a deterministic time-net argument and a short-time continuity estimate for Langevin dynamics make the estimate uniform over $t \in [0, T]$.

4.4 What Does Not Extend

The preceding argument gives only the Lipschitz layer of the pure theorem. In the pure model, Sellke further improves the upper bound $E_\infty(p) + \eta$ to $E_\infty(p) - \delta$. This final refinement uses a gradient lower-bound argument that is special to pure p-spin Hamiltonians and might not be able to be generalized to mixed p-spin models.

The main reason why is the loss of homogenity. In the pure model, $H_{N,p}$ is homogeneous of degree p, so the radial derivative is directly related to the energy. This creates a strong link between the energy level, the radial derivative, and the spherical Hessian geometry. [Sel24b] uses this structure to show that if the trajectory reaches too close to $E_\infty(p)$, then a restarted bounded-temperature Langevin dynamics would force a further increase in energy, contradicting the Lipschitz upper bound. For a mixed Hamiltonian, whihch is inhomogenous, different degrees contribute with different weights. The radial derivative is therefore no longer determined by the energy alone, and the spherical Hessian geometry near a given energy level is not controlled by the same one-parameter relation. Donsequently, the pure-model improvement from $E_\infty(p) + \eta$ to $E_\infty(p) - \delta$ does not directly extend to general mixtures. Therefore, the result in this section should be interpreted as a finite-horizon Lipschitz upper bound. It shows that, for fixed $T < \infty$ and bounded deterministic drift schedule β_t, the soft-approximation and Lipschitz-hardness method prevents annealed Langevin dynamics from exceeding $\mathrm{ALG}(\xi)$ up to an arbitrarily small error. It does not prove that mixed-model simulated annealing reaches $\mathrm{ALG}(\xi)$, nor does it settle the behavior of schedules whose time horizon or temperature scale grows with N.

5 Conclusion

This work reviewed the upper-bound mechanism for Langevin dynamics and simulated annealing in spherical spin glasses. The main question is whether annealed Langevin dynamics can reach the ground state energy or whether it is limited by the lower algorithmic threshold $\mathrm{ALG}(\xi)$. For pure spherical p-spin Hamiltonians, Sellke's theorem shows that bounded-temperature Langevin dynamics is bounded away from the threshold $E_\infty(p)$ from above: for each fixed temperature bound β_0, the trajectory satisfies

$$\sup_{t\in[0,T]} \frac{H_{N,p}(x_t)}{N} \le E_\infty(p) - \delta$$

with high probability, for a positive $\delta = \delta(p, \beta_0)$. The proof has two conceptual parts. First, the hard spherical dynamics is approximated by a soft full-space dynamics with a radial confining potential. This removes the main technical obstacle caused by the projected Brownian noise on the sphere. Second, after conditioning on the Brownian path, the soft endpoint is shown to be a dimension-free Lipschitz function of the disorder. Lipschitz hardness then prevents the endpoint from exceeding the algorithmic threshold. In the pure case this gives the bound $E_\infty(p) + \eta$, and Sellke's additional gradient lower-bound argument improves it to $E_\infty(p) - \delta$. For mixed spherical p-spin Hamiltonians, the soft approximation and Lipschitz reduction are robust. [Sel24b] give the finite-horizon upper bound

$$\sup_{t\in[0,T]} \frac{H_N(x_t)}{N} \le \mathrm{ALG}(\xi) + \eta, \qquad \mathrm{ALG}(\xi) = \int_0^1 \sqrt{\xi''(q)}\,\mathrm{d}q,$$

with high probability. What does not immediately extend is the final pure-model refinement below the threshold. The reason is that pure models have a special homogeneity relation between energy, radial derivative, and Hessian geometry, while mixed models combine several degrees and no longer have this one-parameter structure. Thus the mixed-model result should be viewed as a Lipschitz upper bound, not as a complete characterization of the annealing energy. The matching lower bound, namely whether annealed Langevin dynamics actually reaches $\mathrm{ALG}(\xi)$ for general mixtures, remains the main open direction.

References

[BADG06] Gérard Ben Arous, Amir Dembo, and Alice Guionnet. Cugliandolo-kurchan equations for dynamics of spin-glasses. *Probability theory and related fields*, 136(4):619–660, 2006.

[CK93] Leticia F Cugliandolo and Jorge Kurchan. Analytical solution of the off-equilibrium dynamics of a long-range spin-glass model. *Physical Review Letters*, 71(1):173, 1993.

[CS92] Andrea Crisanti and H-J Sommers. The spherical p-spin interaction spin glass model: the statics. *Zeitschrift für Physik B Condensed Matter*, 87(3):341–354, 1992.

[EAMS21] Ahmed El Alaoui, Andrea Montanari, and Mark Sellke. Optimization of mean-field spin glasses. *The Annals of Probability*, 49(6):2922–2960, 2021.

[HMRW25] Brice Huang, Sidhanth Mohanty, Amit Rajaraman, and David X Wu. Weak poincaré inequalities, simulated annealing, and sampling from spherical spin glasses. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing*, pages 915–923, 2025.

[HS25] Brice Huang and Mark Sellke. Tight lipschitz hardness for optimizing mean field spin glasses. *Communications on Pure and Applied Mathematics*, 78(1):60–119, 2025.

[Sel24a] Mark Sellke. Optimizing mean field spin glasses with external field. *Electronic Journal of Probability*, 29:1–47, 2024.

[Sel24b] Mark Sellke. The threshold energy of low temperature langevin dynamics for pure spherical spin glasses. *Communications on Pure and Applied Mathematics*, 77(11):4065–4099, 2024.

[SK75] David Sherrington and Scott Kirkpatrick. Solvable model of a spin-glass. *Physical review letters*, 35(26):1792, 1975.

[Sub21] Eliran Subag. Following the ground states of full-rsb spherical spin glasses. *Communications on Pure and Applied Mathematics*, 74(5):1021–1044, 2021.